The Meteoritical Society

Meteoritics & Planetary Science 46, Nr 11, 1670–1687 (2011)
doi: 10.1111/j.1945-5100.2011.01255.x

Rock magnetic and paleomagnetic study of the Keurusselkä impact structure, central Finland

Selen RAISKILA[1*], Johanna SALMINEN[1,2], Tiiu ELBRA[1], and Lauri J. PESONEN[1]

[1]Department of Physics, University of Helsinki, P.O. Box 64, 00014 University of Helsinki, Helsinki, Finland
[2]Department of Geology and Geophysics, Yale University, New Haven, Connecticut 06520–8109, USA
*Corresponding author. E-mail: selen.raiskila@helsinki.fi

(Received 25 March 2011; revision accepted 16 August 2011)

Abstract–There are 31 proven impact structures in Fennoscandia—one of the most densely crater-populated areas of the Earth. The recently discovered Keurusselkä impact structure (62°08′ N, 24°37′ E) is located within the Central Finland Granitoid Complex, which formed 1890–1860 Ma ago during the Svecofennian orogeny. It is a deeply eroded complex crater that yields in situ shatter cones with evidence of shock metamorphism, e.g., planar deformation features in quartz. New petrophysical and rock magnetic results of shocked and unshocked target rocks of various lithologies combined with paleomagnetic studies are presented. The suggested central uplift with shatter cones is characterized by increased magnetization and susceptibility. The presence of magnetite and pyrrhotite was observed as carriers for the remanent magnetization. Four different remanent magnetization directions were isolated: (1) a characteristic Svecofennian target rock component A with a mean direction of D = 334.8°, I = 45.6°, α_{95} = 14.9° yielding a pole (Plat = 51.1°, Plon = 241.9°, A_{95} = 15.1°), (2) component B, D = 42.4°, I = 64.1°, α_{95} = 8.4° yielding a pole (Plat = 61.0°, Plon = 129.1°, A_{95} = 10.6°), (3) component C (D = 159.5°, I = 65.4°, α_{95} = 10.7°) yielding a pole (Plat = 21.0°, Plon = 39.3°, A_{95} = 15.6°), and (4) component E (D = 275.5°, I = 62.0°, α_{95} = 14.4°) yielding a pole (Plat = 39.7°, Plon = 314.3°, A_{95} = 19.7°). Components C and E are considered much younger, possibly Neoproterozoic overprints, compared with the components A and B. The pole of component B corresponds with the 1120 Ma pole of Salla diabase dyke and is in agreement with the $^{40}Ar/^{39}Ar$ age of 1140 Ma from a pseudotachylitic breccia vein in a central part of the structure. Therefore, component B could be related to the impact, and thus represent the impact age.

INTRODUCTION

Approximately 178 impact structures have been presently confirmed on the Earth (Earth Impact Database 2011). Fennoscandia is statistically one of the most densely crater-populated parts of the Earth (Henkel and Pesonen 1992; Abels et al. 2002; Dypvik et al. 2008) as most of the craters are exposed. Altogether, 31 impact structures, with ages from approximately 2400 Ma to approximately 0.004 Ma, have been discovered due to the high level of research activity (Fig. 1A). Studies of magnetic properties of several Fennoscandian impact structures have been performed successfully, e.g., Lappajärvi (Pesonen et al. 1992; Pipping and Lehtinen 1992) and Karikkoselkä (Pesonen et al. 1999) in Finland; Kärdla in Estonia (Plado et al. 1996); Jänisjärvi in Russian Karelia (Salminen et al. 2006); Lockne, Tvären (Törnberg and Sturkell 2005), and Siljan (Reimold et al. 2005) in Sweden; and Mjølnir in Barents Sea (Tsikalas et al. 1998).

The Keurusselkä is the eleventh impact structure discovered in Finland (Hietala and Moilanen 2004, 2007; Pesonen et al. 2004; Ruotsalainen et al. 2006). The structure (center point 62°08′ N, 24°37′ E) is located in the Paleoproterozoic (Svecofennian) granite terrain in central Finland. It is a deeply eroded structure, and it

© The Meteoritical Society, 2011.

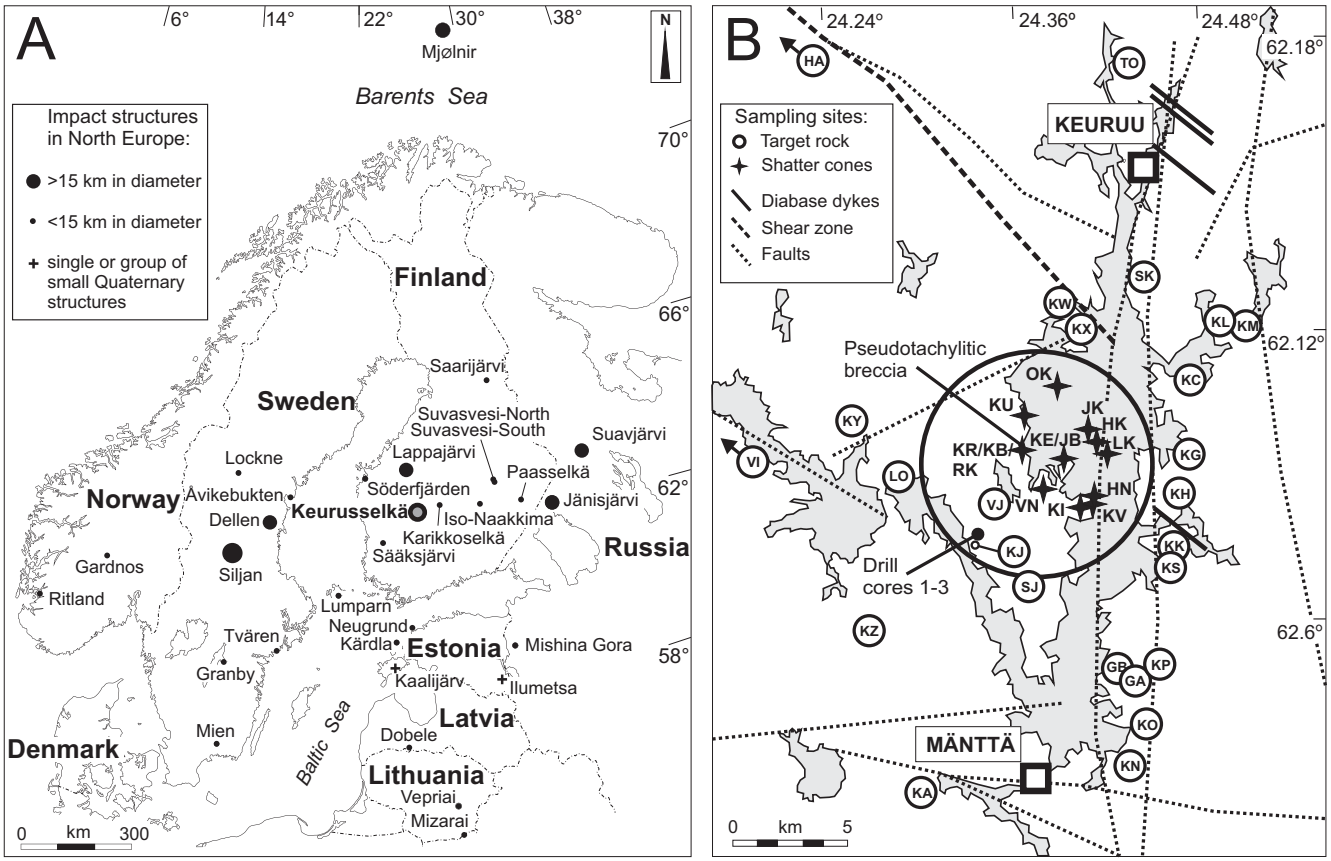


Fig. 1. A) Proven and considered impact structures in North Europe (map modified from Abels et al. 2002) and B) the Keurusselkä sampling sites. Solid ring marks the area of proposed central uplift. Sites with an arrow are sampled outside the map region along a shear zone/faults (map modified after Nironen 2003).

yields in situ rocks with distinct evidences of shock metamorphism such as shatter cones containing planar deformation features (PDFs) (Ferrière et al. 2010).

For this study, we present field mapping with impact evidences (shatter cones, impact breccia, and PDFs). Then, we focus on petrophysical and rock magnetic results for identifying generated changes in the various lithologies, impact shock features, and carriers of remanent magnetization. After that, we present paleomagnetic results of shocked target rocks to date the impact event.

GEOLOGICAL SETTING

The Keurusselkä impact structure is situated within the Central Finland Granitoid Complex (hereafter called CFGC) (Nironen 2003). CFGC was formed 1890–1860 Ma ago during the peak phase of the Svecofennian orogeny, where the growing Svecofennian island arc system collided against the Archaean continent. Continental deformation, metamorphism and crust forming magmatism occurred at approximately 1900–1800 Ma. The granitoids with biotite, hornblende, titanite, apatite, magnetite, and zircon (Kähkönen 2005) are predominant rocks in the CFGC together with schists and gneisses. Arc-type volcanic rocks, which are metamorphosed at high-T and low-P amphibolite facies, characterize also this part of the Paleoproterozoic crust. According to Kähkönen (2005), the Svecofennian granitic bedrock resulted from remelting of older approximately 2100 Ma crust of several microcontinents. Based on ε_{Nd} (Nd isotopic ratio) values the Hirsilä belt, part of the Tampere schist belt, which reaches the southern parts of CFGC, includes an Archaean component as well as abundant approximately 2000 Ma detritus from the suggested microcontinents. While a majority of the Hirsilä belt rocks represent a volcanic arc, the gneisses and migmatites are associated with the subduction complex. Supracrustal rocks of Svecofennian domain are, in many cases, separated by faults and intrusions dividing them into minor belts. Tectonic and metamorphic evolution of Tampere belt caused also complex folding with several phases (Kähkönen 2005).

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The regional area of the Keurusselkä impact structure consists of granitoids, volcanic rocks, and gneisses, which are mainly exposed close to the shoreline of Lake Keurusselkä (Fig. 1B). The Svecofennian gabbro intrusions are exposed on the southeast side of the Keurusselkä structure. The Keuruu diabase dykes (1880 Ma) are cutting tonalitic rocks in the northern parts of the structure (Puranen et al. 1992). Granites reveal two petrologically distinctive types similar to the target rock in the Karikkoselkä structure, which is located 30 km to the east from the Keurusselkä (Pesonen et al. 1999): reddish granite with salmon pink K-feldspar and grayish granite, with high biotite content. Overall, the target rock in the Keurusselkä region is petrologically variable, and granitoids change from fine-grained to coarse-grained and porphyritic. According to database of Geological Survey of Finland (GSF), U-Pb (Zr) ages of the granitoids in the Keurusselkä region show crystallization age of 1883 Ma (Keuruu diorite), 1882 Ma (Karikkoselkä granite and gabbros in CFGC), and 1880 Ma (Multia granite; northern parts of the Keurusselkä), while granites, away from the Keurusselkä region, show U-Pd (Zr) ages of 1886 Ma. Volcanic rocks in CFGC are somewhat older, approximately 1900 Ma. The Keurusselkä target rocks show also significant alteration of minerals due to postglacial uplift and postshock processes. Three major shear zones cross, which are probably related to late Svecofennian events (Nironen 2003), cut the Keurusselkä structure from NW, WE, and NS directions. The NW fracture zone is clearly recognized as a weakly magnetic lineament in the aeromagnetic map based on the data of GSF (Fig. 2).

The Keurusselkä impact structure is named after the lake Keurusselkä, which is considered to be partly the depression of the impact structure. Due to erosion and glaciations, typical for Finnish lakes, the present shape of the lake is elongated, and it is generally shallow, mainly between 2 and 10 m. Based on studies of Keuruu diabase dykes and their contacts against gabbros and granodiorites, up to 10 km (Palmu 1982) of bedrock has been eroded since the formation of diabase dykes at 1880 Ma ago.

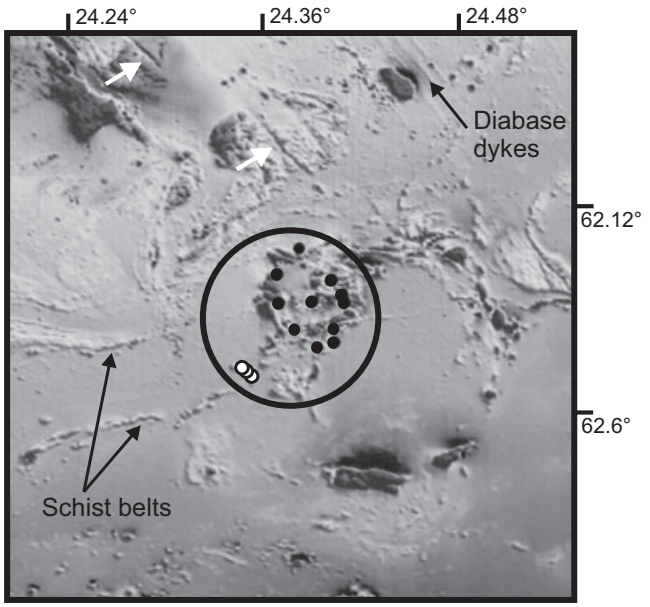


Fig. 2. Aeromagnetic gradient map of the Keurusselkä region. Shatter cone sites are marked as black dots, and the three drill cores with white dots. Central uplift with shatter cones has circular anomaly of increased magnetization. Shear zone from the NW direction is marked with white arrows (map by Geological Survey of Finland, H. Leväniemi).

SAMPLE COLLECTION AND METHODS

During the field seasons in 2004–2010, oriented hand and drilled samples were collected (Fig. 1B). The orientation was done using magnetic and/or sun compasses. The total of 141 hand or portable drill core samples were collected and from each sample, 2–5 standard cylindrical specimens (2.54 × 2.2 cm) were prepared. Most of the measurements were carried out at the Solid Earth Geophysics Laboratory (University of Helsinki). Density, magnetic susceptibility, and natural remanent magnetization (NRM) were measured for minimum of two specimens from each sample. Porosity was measured for 87 samples from 23 sites. The density and porosity measurements were conducted using Archimedean method based on weighting water-saturated and oven-dried samples in both water and air (Kivekäs 1993). Petrophysical measurements of magnetic susceptibility and NRM were conducted using the RISTO-5 instrument in conjunction with kappabridge (operating frequency of 1025 Hz and DC field intensity 48 A m^{-1}). Susceptibility as a function of temperature (Curie temperature) was measured with AGICO's KLY-3S kappabridge by heating the specimens in argon gas environment from room temperature up to 700 °C and cooling back to 40 °C. The magnetic hysteresis, for determination of domain states of the magnetic carriers, and first order reversal curve (FORC) diagrams were measured using a Princeton Measurement Corporation MicroMag™3900 Vibrating Sample Magnetometer (VSM). The thermomagnetic and hysteresis properties were measured using powdered bulk specimens. The FORC data were processed using a code provided by Winklhofer and Zimanyi (2006).

Paleomagnetic measurements were carried out using 2G Superconducting Rock Magnetometer (SQUID). Samples were demagnetized using alternating field (AF) up to 160 mT. Thermal demagnetizations (temperatures

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

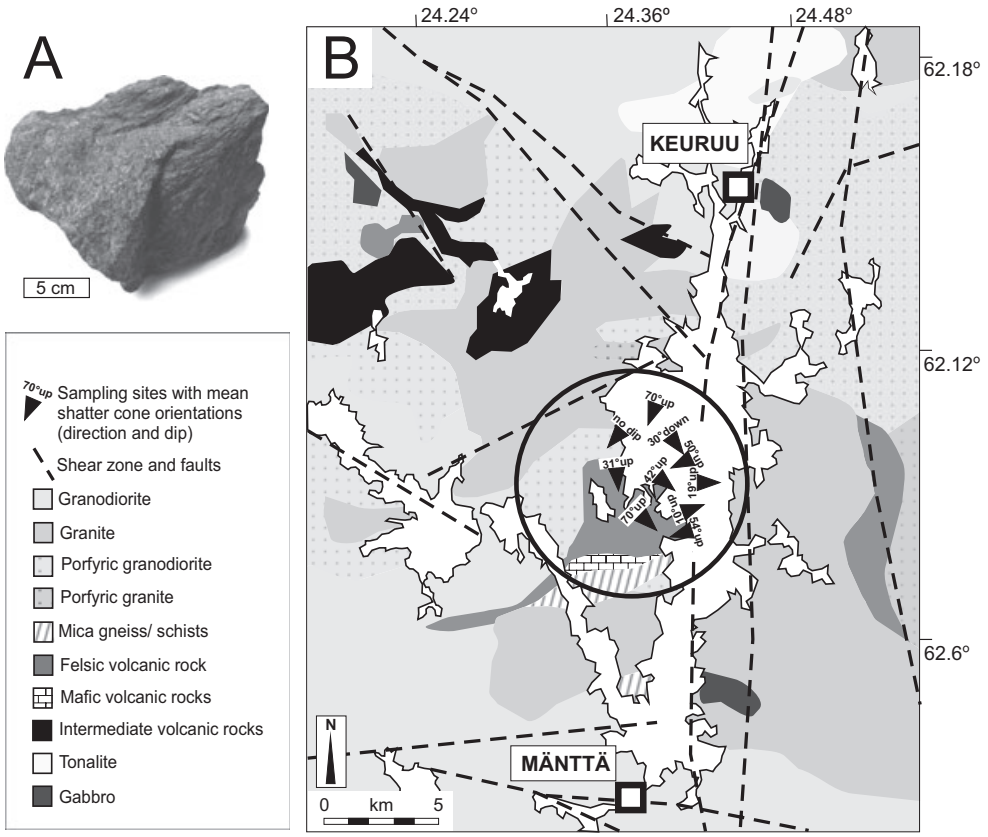


Fig. 3. A) Shatter cone boulder of metavolcanic rock and B) geological map showing shatter cone sites with average shatter cone orientations (geological map modified after Nironen 2003).

up to 700 °C) were done using the Magnetic Measurements MMTD-60 oven and KLY-3S for monitoring the susceptibility. Additional set of specimens were thermally demagnetized (temperatures up to 600 °C) in the paleomagnetism laboratory of the Department of Geology and Geophysics at Yale University in a magnetic shielded room, using nitrogen atmosphere to prevent changes due to oxidation during the heating. To confirm the magnetic carriers, Scanning electron microscope (SEM; JEOL JSM-5900LV) analysis was performed in collaboration with the GSF for selected surface-polished specimens from the shatter cone sites KE and KB. The thin-sections of selected specimens were prepared in collaboration with the Division of Geology, University of Helsinki.

The demagnetization results were analyzed using stereographic projections, demagnetization decay curves, and orthogonal plots (Zijderveld 1967; Leino 1991). The components with minimum of four demagnetization steps were accepted for principal component analysis (Kirschvink 1980) with angular deviation (MAD) less than 6°. Mean remanence directions were calculated using Fisher (1953) statistics.

IMPACT FEATURES

Shatter Cones, Pseudotachylitic and Monomictic Breccia

The first indication of the Keurusselkä impact structure was the discovery of in situ shatter cones in metavolcanic and granitic rocks (Hietala and Moilanen 2004, 2007). In Fig. 1B are presented only the sites (Ferrière et al. 2010; this study) with clear and recognizable in situ shatter cones. Boulders of shatter cones were found in several locations near the in situ ones. In situ shatter cones are mainly 15 cm long and 5–10 cm wide displaying pointing apices (Fig. 3). The cones are well developed, although they occasionally appear only as conical, striated surfaces. Clearly recognized in situ shatter cones are exposed in an area of diameter of 5 km. However, Hietala and Moilanen (2007) reported 65 outcrops in approximately 14 km wide area including also shatter cone-like features, which are most possibly only slicken sides as the striations lack the clear pointing apices.

In general, the allochthonous breccia lens with fractured and shocked rocks is the main macroscopic

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

feature present on impact craters (Melosh 1989). In the Keurusselkä area, the bedrock is strongly fractured, which was observed during the field studies, and radial impact fracturing patterns in outcrops were distinguished from fracture zone directions described by Nironen (2003) and the common linear glacial striations detected across Finland. Quantity of shatter cones indicates also a major shock effect. These shatter cones are considered to present the central uplift of the structure as they appear in a circular area close to the presumed center point of the impact (Ferrière et al. 2010). Figure 3 demonstrates that a majority of shatter cones are pointing upward, and have a tendency to point away from the lake. The steep upward direction and the lack of well-defined shatter cone intersection points may suggest that the shatter cones are peripherically located around the central uplift (Dr. A. Deutsch, personal communication).

Large impact breccia deposits or impact melt, however, were not found, which indicates that Keurusselkä is indeed deeply eroded, and possibly much older compared with the other Finnish impact structures (Dypvik et al. 2008). Schmieder et al. (2009) reported a discovery of a thin (approximately 2 cm) pseudotachylitic breccia vein (site RK in Fig. 1B) cutting shatter cones on the western shore of the lake Keurusselkä. This pseudotachylitic breccia was dated using $^{40}Ar/^{39}Ar$-method and an average plateau age of 1140 ± 6 Ma was obtained (Schmieder et al. 2009). However, pseudotachylitic breccia veins are quite common in geological environments, and they are usually also associated with tectonic events (Reimold 1998). Yet, the discovery of weak shock features (PFs) in quartz (Schmieder et al. 2009) could point to the impact origin for the pseudotachylitic breccia vein and suggest a Mesoproterozoic age for the impact event.

In 2009, our group made an observation of narrow monomictic breccia veins from one of the drill cores drilled close to the central uplift area (Fig. 1B). The breccia has a resemblance to the impact-related lithic parautochtonous impact breccias generally formed beneath the crater floor in relatively low pressures (French and Koeberl 2010). The angular and slightly roundish gneiss clasts are embedded in a fine-grained dark gray matrix of the same material. Thin section analysis of these breccia samples showed hydrothermal alteration and chloritization in minerals (Raiskila et al. 2011). Different sulfides like chalcopyrite, pyrite, and trace amounts of pyrrhotite were detected. PDFs in quartz were not found, but few small (less than 1 mm) optically isotropic areas were noticed. Microprobe studies revealed these areas as very fine-grained clay minerals (chlorite, illite, smectite-group). Based on their appearance, it seems that they have originated from glass. The geochemical analysis is currently ongoing, and results will be reported in the near future.

Planar Deformation Features

The shock produced Planar Deformation Features (PDFs) in quartz, and feldspar is a clear evidence of a hypervelocity impact, as they are distinct from nonimpact deformation features (French and Koeberl 2010). PDFs are also critical in recognizing the deeply eroded impact structures, such as the Keurusselkä structure, lacking impact lithologies. Thin-section analyses were performed in 2007 for sites KE and KB, with well-developed shatter cones, to study shock features in minerals. Only one of the samples (KB) showed weak decorated Planar Features (PF), which are not regarded as unambiguous evidence of shock metamorphism (French 1998). Recently, Ferrière et al. (2010) reported new results from selected samples (sites KE, VN) using universal-stage measurement method (Ferrière et al. 2009). Sites KE and VN revealed clear impact-related shock features, such as PDFs and characteristic mineralogical changes due to the impact event. Additional thin sections, from sites KE and VN, confirmed the abundance of shock minerals for in situ shatter cone samples. Examination of these thin sections showed that quartz grains with PDFs are mainly granulated. Overall, PDFs were found in a variety of minerals including quartz, plagioclase, and also apatite; specimens from site KE contain weak PDFs in plagioclase with set of PFs and kink bands in biotite grains. Thin-section sample (site KE) showed similar features to those of thin-sections of the Lappajärvi impact rocks (Lehtinen 1976) and the Karikkoselkä borehole samples (Pesonen et al. 1999). Sample from site VN contains somewhat larger quartz grains with PDFs.

Geophysical Signatures

Magnetic crystalline basement could be exposed at deeply eroded structures, and these localized anomalies have usually large amplitude and occur at or near the crater center (Pilkington and Grieve 1992). In the Keurusselkä structure, the shatter cones within crystalline rocks are exposed in an area of 19 km^2, which shows distinctively higher magnetization features on aeromagnetic map (Fig. 2). This anomaly is interpreted to form the circular central uplift of the structure. On the eastern side of the central uplift is also a recognizable magnetic anomaly forming a partly seen ring around the central uplift area. The magnetic pattern in the Keurusselkä area shows linear and folded schist belts and roundish granitoid intrusions, which are seen as magnetic lows. Raiskila et al. (2011) proposed that

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

results from the drill core samples and the Keurusselkä shatter cones imply magnetic enhancement of the schist belt at the central uplift due to the impact.

Gravity data of the Keurusselkä region also show a circular gravity minimum around the central uplift, which is identified by gravity high. Chain of positive gravity anomalies with a diameter of 25 km is recognized around the central uplift as possible traces of the rim.

RESULTS

Petrophysical Properties

Petrophysical results of various lithologies are presented in Table 1 and are compared with the GSF database records of the GFGC (Säävuori and Hänninen 1997). As the target rocks in the crystalline basement consist mainly of granitoids, the rocks were separated concerning the degree of metamorphoses.

The magnetic susceptibility of the analyzed specimens varies widely. On average, the susceptibilities are between 200 and 500×10^{-6} SI (Fig. 4). However, the main shatter cone sites, which are considered to represent the central uplift, reveal higher susceptibilities $> 1000 \times 10^{-6}$ SI. The Koenigsberger's ratio (Q ratio) is generally low, less than 1. Similar low Q ratios have been observed also for other impact structures in CFGC: for the Karikkoselkä (Pesonen et al. 1999) and for the Lappajärvi impact structures (Pesonen et al. 1992). However, the two sites, KC and KG, show higher Q ratios (> 10) due to higher NRM values. As these sites do not show any shock features, the increased NRM intensities are possibly caused by a lightning strike.

Magnetic Mineralogy

Magnetic properties (Table 2) of the Keurusselkä rocks were determined to investigate the composition and grain size distribution of magnetic minerals to study the stability of natural remanent magnetization for paleomagnetic purposes and to seek correlations between shocked and unshocked rocks caused by the impact.

According to thermomagnetic results (Fig. 5 and Table 2), most of the samples show two or three magnetic phases. The first weak phase occurs around 120°–150°. The second phase, T_C, with higher coercivity, occurs at approximately 330°, indicating most likely the presence of pyrrhotite. The third phase with Curie temperatures of approximately 560–580 °C and occurrence of Hopkinson peaks indicates almost pure magnetite of single-domain (SD) and/or pseudo-single-domain (PSD) type. Some thermomagnetic curves show also possible weak traces of hematite around 680 °C. Most samples indicated irreversible behavior of heating and cooling cycle, which is typical for magnetite, and indicates instability of ferromagnetic minerals with formation of new magnetic phases (Dunlop and Özdemir 1997). Irreversibility could also be caused by maghemization during the heating.

Magnetic hysteresis was measured to characterize the domain states of ferromagnetic minerals. Most of the specimens show low coercivity values (< 0.1 T, maximum 0.4 T), which indicate magnetite (O'Reilly 1984; Dekkers 1988). Hysteresis measurements from the site (KP) show also saturation of fields up to 0.9 T, indicating the presence of higher coercivity minerals. However, the Curie temperature of 560 °C for this sample indicates only the presence of magnetite.

First Order Reversal Curve (FORC) diagrams are used to characterize and identify the contributions of magnetic grains (Roberts et al. 2000) (Fig. 6). The magnetic mineral fractions indicate mostly pseudo-single domain (PSD) magnetite, but the influence of pyrrhotite and possible hematite or maghemite is observed from half of the samples with different magnetic phases shown as bent contours (Wehland et al. 2005). Single-domain (SD) behavior was identified for one target rock site. The FORC diagrams show mostly symmetrical contours with vertical spreading. However, few specimens reveal also asymmetry in negative values indicating increased grain sizes. SEM (Scanning Electron Microscope) analysis of polished specimen surfaces from the central shatter cone sites, KB and KE, proves the presence of magnetite with various grain sizes. Fracturing of the magnetite grains indicates partial alteration to hematite (Fig. 7). Traces of SD or PSD type magnetite or pyrrhotite were also notified from the selected specimens. These fine-grained magnetic minerals can efficiently carry remanent magnetization.

Paleomagnetic Results

Paleomagnetic results are shown in Table 3. Both alternating field (AF) up to 160 mT and thermal treatments (TH) up to 600 °C/700 °C were conducted for selected specimens to isolate the remanenent magnetization components (Figs. 8 and 9). Due to weak intensities and often also unstable demagnetization behavior, only less than half of the selected and demagnetized specimens (96 out of 232) showed stable remanence directions. Nevertheless, these demagnetization results provide important geological information over the Keurusselkä region.

Some of the target granitoids do not carry any stable remanent magnetization, and the demagnetization behavior is erratic (sites KG, KN and KV). The AF demagnetization shows typically low or intermediate (5–10 mT up to 50 mT) coercivities. However, unshocked

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Petrophysical properties of Keurusselkä target rocks and shatter cones.

Rock type	Site	*B*N/n*	*s* (km)	*d* (kgm^{-3})	*P* (%)	*k* (10^{-6} SI)	NRM (mAm^{-1})	J_{20}/J_0	*Q*
Target rock									
Gabbro	GA	5/9	8.7	3160	0.4	45,800	2480	0.4	1.2
	GB	5/10	8.5	3098	0.3	5180	233	0.2	0.8
Mean		2*/10/19		3130 (2900)	0.4	25,500 (10,300)	1360 (560)	0.3	1.0 (2.3)
Tonalite	TO	1*/5/5	18	2600 (2730)	1.1	5300 (3120)	40 (106)	0.2	0.2 (2.3)
Granite	KO	5/6	11	2730		350	9	0.4	0.7
	SK	5/9	9.8	2730		350	6	0.6	0.4
	KP	5/10	8.8	2630		1640	15	0.2	0.2
	KC	1/8	8.1	2700	0.5	200	180	0.4	24
	KX	4/24	7.2	2640		1090	35	0.5	3.3
	KG	5/9	6.4	2690	0.6	190	560	0.7	63
	KK	5/34	6.3	2660		220	33	0.6	5.2
	KS	5/9	6.2	2700		240	3	0.6	0.3
	KH	4/12	5.6	2650	0.7	140	25	0.8	4.9
	SJ	2/10	3.9	2580		170	5	0.5	0.9
Mean		10*/36/122		2670 (2640)	0.6	460 (240)	87 (54)	0.5	10 (11)
Granodiorite	VI	5/7	22	2710		280	14	0.5	1.3
	KA	2/6	16	2700		180	5	1.0	0.9
	KN	4/4	13	2810		380	3	0.9	0.2
	KM	3/23	11	2720	0.5	310	10	0.9	0.9
	KL	5/8	11	2710		290	9	0.9	0.8
	KZ	2/11	9.8	2720	1.0	260	38	0.5	3.8
	KY	3/26	9.2	2650		340	35	0.6	0.5
	KW	2/10	8.1	2720	1.0	270	22	0.5	2.9
	LO	2/6	6.0	2650		200	2	0.9	0.3
Mean		9*/23/94		2710 (2675)	0.8	280 (290)	15 (53)	0.7	1.3 (6.0)
Gneiss	HA	5/11	25	2700		160	3	0.4	0.6
	KJ	5/10	4.2	2730		340	9	0.6	0.6
	VJ	7/11	2.1	2600		400	4	0.3	0.2
Mean		3*/12/21		2680 (2701)		300 (322)	5 (64)	0.4	0.5 (5.8)
Metavolc. rock				(2680)		(8030)	(1300)		(8.9)
Shatter cones									
Granite	KB	5/34	1.8	2810	0.9	12,000	390	0.1	0.7
	RK	4/16	1.8	2740		5100	110	0.2	0.4
	KR	6/17	1.8	2830	0.9	3800	45	0.1	0.3
Mean		3*/15/67		2790	0.9	6700	180	0.1	0.5
Granodiorite	KU	1*/5/5	3.3	2790		410	3	0.7	0.2
Gneiss	OK	5/22	4.5	2670	0.7	220	11	0.4	1.5
	LK	2/10	3.1	2680	0.6	17,000	407	0.1	0.6
	HN	1/5	2.1	2620		120	17	0.2	2.8
	KE	4/34	1.9	2730	0.8	25,000	614	0.2	0.6
	KI	1/2	1.7	2600		2500	39	0.3	0.4
Mean		5*/13/73		2660	0.7	8900	220	0.2	1.2
Metavolc.	JK	4/15	3.4	2690	0.6	32,000	730	0.1	0.6
rock	HK	5/19	3.3	2760	0.7	31,000	840	0.1	0.7
	KV	6/26	2.1	2670	0.9	190	5	0.5	0.6
	JB	3/20	1.9	2740		37,000	690	0.2	0.5
	VN	3/14	0	2580		460	8	0.6	0.4
Mean		5*/21/94		2690	0.7	20,100	450	0.3	0.6
Pseudotach. breccia	RK2	1*/1/1	1.8	2780		670	8	0.3	0.3

Notes: *B** = number of sites, *N* = the number of hand samples or sample cores, and *n* = the number of specimens. Statistical calculations were made using specimen level; *s* = the distance from the crater center; *d* = density; *P* = porosity; *k* = susceptibility; *J* = intensity of natural remanent magnetization; J_{20}/J_0 = intensity of magnetization hardness; and *Q* = the Koenigsberger's ratio calculated for a field of 50 μT. Values in parentheses are reference values for different lithologies outside the impact region from Geological Survey of Finland (GSF) database.

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

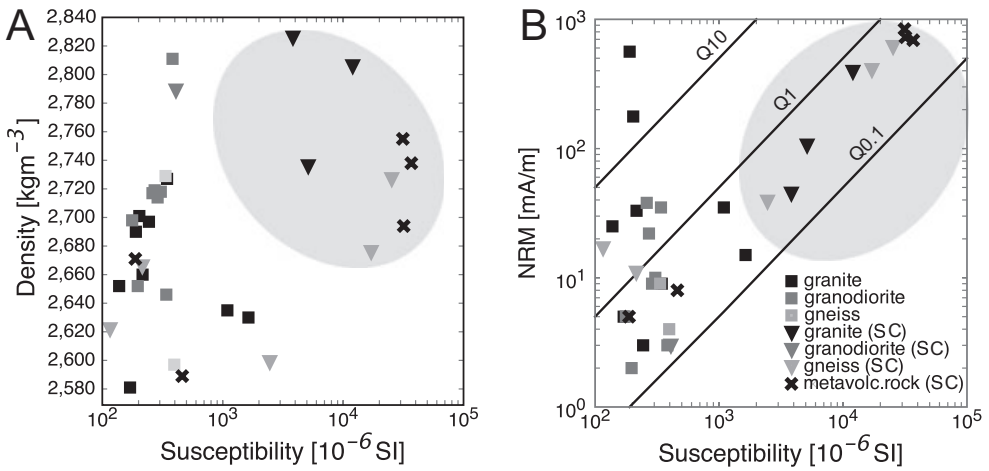


Fig. 4. A) Density versus susceptibility and B) Natural remanent magnetization versus susceptibility of different lithologies (with equal Q-ratio lines). A majority of the shatter cone (SC) specimens (gray area) show enhanced susceptibilities.

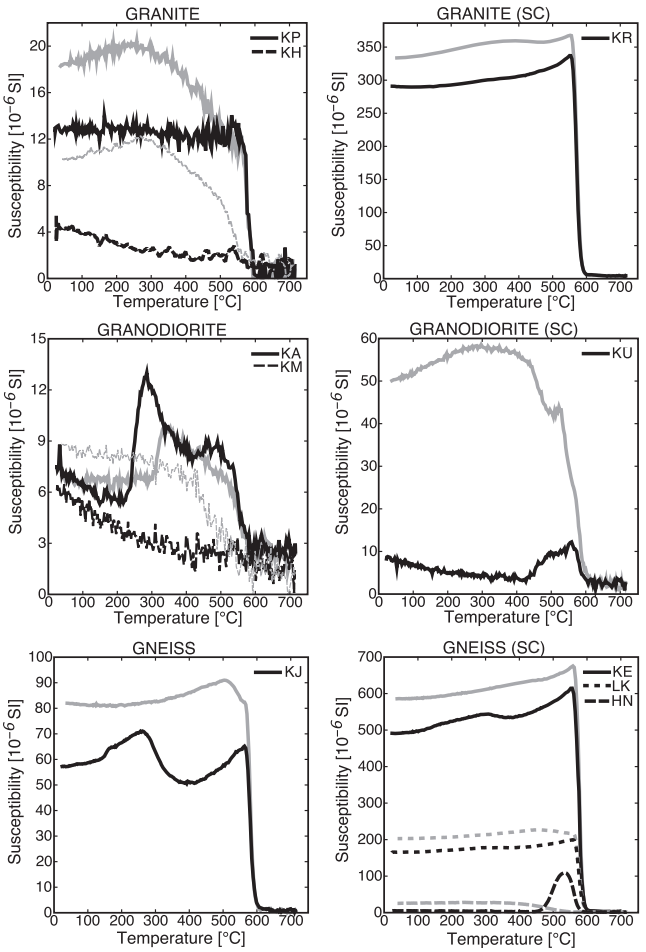


Fig. 5. Selected thermomagnetic curves in Argon gas for unshocked target rock and shatter cones of different lithologies. Black color marks the heating curves, and gray color cooling curves.

or weakly shocked granodiorite and granite specimens from the sites KH, KL, KM, KN and SJ show higher values for the coercivity. NRM of some specimen did not decay during the AF demagnetization (up to 160 mT).

During thermal demagnetization, the mineralogical changes often occurred before or after 330 °C, indicating formation of new magnetic phases such as pyrrhotite. Therefore, a set of specimens were thermally demagnetized in magnetically shielded room and in nitrogen gas environment. For these specimens, liquid nitrogen was used to remove the possible viscous magnetization in MD grains before the demagnetization.

As the studied samples present the rock formation itself and the deeply eroded crater basement, we were not able to use paleomagnetic field tests for verifying the primary nature of isolated remanent magnetization directions.

Component A

NNW upward intermediate component A is observed for seven sites including three (KB, KE, and OK) shatter cone sites. This component is best preserved on sites KL and KM approx. 11 km away from the impact center. It is especially observed during the thermal demagnetization showing blocking temperatures between 200 °C and 540 °C. The same remanent magnetization direction with intermediate coercivity is also isolated during the AF treatment for these sites. This direction is also clearly shown on sites SK and VI, which are located 10–20 km away from the impact center and close to the faults, although component separation on site VI was unsuccessful.

Specimens with component A have nearly constant declinations, but the inclinations show variation of

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Rock magnetic properties of Keurusselkä target rocks and shatter cones.

Rock type	Site	s (km)	M_r (mAm^2kg^{-1})	M_s (mAm^2kg^{-1})	H_{cr} (mT)	H_c (mT)	M_r/M_s	H_{cr}/H_c	T_C (°C)
Target rock									
Gabbro	GA	8.7	25	600	21	4.3	0.042	4.9	330,570
	GB	8.5	9.0	580	9.3	1.4	0.016	6.6	330,580
Mean			17	590	15	2.9	0.029	5.8	
Tonalite	TO	18.4	9.8	480	25	2.3	0.02	10.77	560
Granite	KO	11	0.10	0.83	110	15	0.12	7.3	540
	SK	9.8	2.3	2.6	26	5.7	0.88	4.6	n.d.
	KP	8.8	1.4	32	23	3.4	0.044	6.8	570
	KC	8.1	0.043	0.50	37	10	0.086	3.7	n.d.
	KX	7.2	0.11	0.82	150	15	0.13	10	n.d.
	KG	6.4	0.022	0.32	180	170	0.069	1.1	n.d.
	KK	6.3	0.040	0.51	33	270	0.078	0.12	n.d.
	KS	6.2	0.057	1.5	60	4.3	0.038	14	n.d.
	KH	5.6	0.02	1.3	610	5.8	0.015	105	n.d.
	SJ	3.9	1.0	5.6	170	27	0.18	6.3	520
Mean			0.51	4.6	140	53	0.16	16	
Granodiorite	VI	22	0.20	1.1	35	7.5	0.18	4.7	n.d.
	KA	16	6.2	13	102	98	0.48	1.0	570
	KN	13	0.14	0.62	75	29	0.23	2.6	570
	KM	11	0.071	0.93	69	9.0	0.076	7.7	n.d.
	KL	11	0.096	1.4	46	8.9	0.069	5.2	n.d.
	KZ	9.8	0.085	1.0	47	7.6	0.085	6.2	n.d.
	KY	9.2	0.18	3.1	51	6.8	0.058	7.5	n.d.
	KW	8.1	0.15	1.4	38	6.8	0.10	5.6	n.d.
	LO	6.0	0.089	0.48	62	23	0.19	2.7	n.d.
Mean			0.80	2.6	58	22	0.16	4.8	
Gneiss	HA	25	0.07	3.4	57	18	0.020	3.2	540
	KJ	4.2	1.6	3.7	85	63	0.43	1.3	330,570
	VJ	2.1	0.43	7.4	41	6.4	0.058	6.4	n.d.
Mean			0.7	4.8	61	29	0.17	3.6	
Shatter cones									
Granite	KB	1.8	32	560	21	6.0	0.057	3.5	580
	KR	1.8	18	390	20	5.4	0.046	3.7	580
	RK	1.8	0.15	2.3	38	6.6	0.065	5.8	580
Mean			17	320	26	6.0	0.056	4.3	
Granodiorite	KU	3.3	0.81	2.4	12	46	0.34	0.25	560
Gneiss	OK	4.5	0.43	3.9	8.9	3.5	0.11	2.5	330
	LK	3.1	20	430	24	5.4	0.047	4.4	330,580
	HN	2.1	6.0	13	24	19	0.46	1.3	560
	KE	1.9	79	1200	27	7.4	0.066	3.6	330,580
	KI	1.7	11	120	64	11	0.092	5.8	330,570
Mean			23	350	30	9.3	0.16	3.5	
Metavolcanic rock	JK	3.4	45	1100	16	4.2	0.041	3.8	330,570
	HK	3.3	98	2100	16	4.2	0.047	3.8	330,560
	KV	2.1	0.042	0.38	63	13	0.11	4.8	n.d.
	JB	1.9	270	3200	37	9.7	0.084	3.8	330,570
	VN	0	3.9	30	52	11	0.13	4.7	540
			83	1300	37	8.4	0.08	4.2	

Notes: s = the distance from the crater center; M_r = saturation remanent magnetization; M_s = saturation magnetization; H_{cr} = coercivity of remanence; H_c = coercive field; T_c = Curie temperature; n.d. = no data.

30–40°. The mean A direction is a typical Svecofennian direction (D = 334.8, I = 45.6°, α_{95} = 14.9°) of remanent magnetization, which yields a paleomagnetic pole Plat = 51.1° and Plong = 241.9° (A_{95} = 15.1°) (Figs. 8 and 10). This type of direction is observed also in numerous locations within the Svecofennian terrain in Finland and Sweden (Pesonen et al. 1989, 1991). Moreover, it is strikingly similar to direction isolated from unshocked granites near the Karikkoselkä impact structure (Pesonen et al. 1999).

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

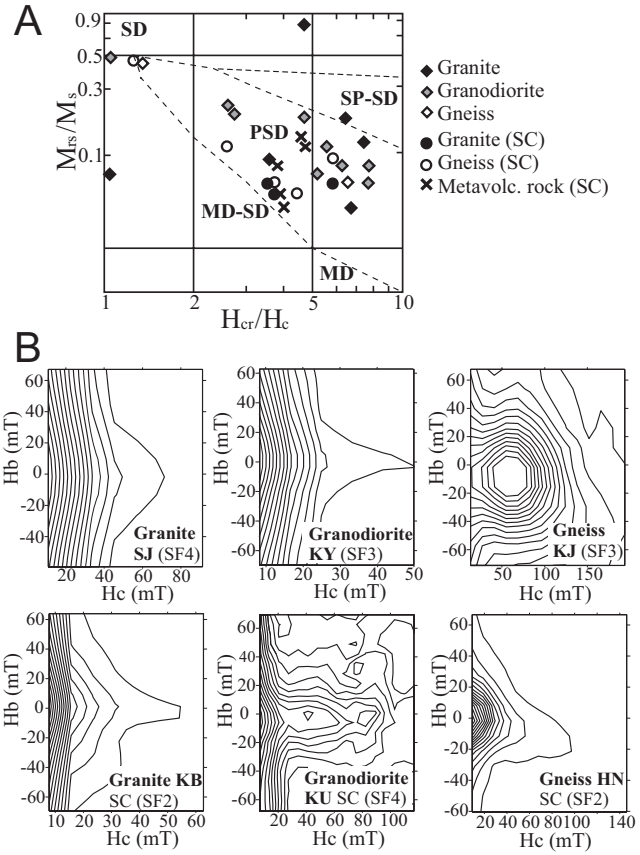


Fig. 6. The distribution of magnetic minerals. A) Day-plot and B) First-order reversal curve diagrams (FORC) for various lithologies of the target rock and shatter cones (SC). SF indicates the used smoothing factor.

Component B

NW upward intermediate inclination component B is isolated for nine sites occasionally occurring in the same samples with component A. It is best preserved in the sites with shatter cones (VN, JK, LK, OK, KB, and KR) showing mostly stable magnetizations and low-to-intermediate coercivities (10 mT up to 40 mT) during the AF treatment. In some cases, the components are isolated even in coercivities below 10 mT, although the direction could be viscose. Few specimens from sites VN, JB, and KB showed also higher coercivities (>100 mT). Blocking temperature ranges between 150 °C and 580 °C. The remanent magnetization of specimens holding the component B often shows steep inclination, while declination is generally more widely distributed (25–55). Especially half of the specimens from site VN, with peak impact pressures around 20 GPa (Ferrière et al. 2010), show directions with steep inclination close to the direction of the present earth field (PEF) of the sampling site (D = 7° and I = 73°).

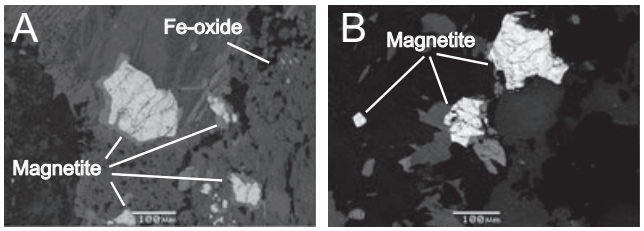


Fig. 7. Scanning electron microscope (SEM) images of magnetic minerals in selected shatter cone specimens. A) Fractured MD type magnetite grains with iron oxide from site KB (granite). B) Magnetite of variable grain sizes from site KE (gneiss).

We were able to separate a remanent magnetization component B (D = 42.4°, I = 64.1°, α_{95} = 8.4°) yielding a pole Plat = 61.0°, Plon = 129.1° (A_{95} = 10.6°) for impact-affected basement samples (Figs. 8 and 10).

Components C and E

Components C and E are isolated from outside the impact region from site HA (shear zone) showing direction C, although component separation was unsuccesful. Samples from site VI (fault) show only E direction. Both of these sites lack the primary Svecofennian direction clearly indicating remagnetization. Interestingly, four shatter cone sites (KE, RK, HK, and OK) show both components.

The mean direction for C is D = 159.5°, I = 65.4°, α_{95} = 10.7° with a pole Plat = 21.0°, Plon = 39.3° (A_{95} = 15.6°) and for E is D = 275.5°, I = 62.0°, α_{95} = 14.4° with a pole Plat = 39.7°, Plon = 314.3 (A_{95} = 19.7°) (Figs. 9 and 10).

DISCUSSION

Petrophysical Properties

Petrophysical properties of the Keurusselkä impact structure are generally typical for the granitic bedrock (CFGC). However, crystalline rocks with shatter cones within the central parts of the structure show enhanced magnetic susceptibilities, which are interpreted to be associated with the central uplift. Increased susceptibilities in Keurusselkä structure could be a result of modification of magnetic carriers due to high P-T-conditions during the impact and later in hydrothermal processes. Hydrothermal activity has been significant after the impact event and it has caused fluid inclusion trails decorating the originally amorphous planar deformation features (PDFs) in quartz (Ferrière et al. 2010). Linear and folded schist belts dominate the overall

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Paleomagnetic results of Keurusselkä target rocks and shatter cones.

Rock type	Site	*s* (km)	*B*/N/n*	*D* (°)	*I* (°)	*k*	A_{95} (°)	Plat (N°)	Plon (E°)	dp (°)	dm (°)	A_{95} (°)	*S* (°)
Comp. A													
Gabbro	GA	8.7	3/2	322.6	15.9			29.5	248.4				
Granite	SK	9.8	2/2	348.5	29.0			42.6	219.8				
Granodiorite	KM	10.8	3/9	333.7	50.9	491.4	5.6	55.1	246.0	5.1	7.5	6.2	3.7
	KL	10.5	4/4	330.6	42.7	69.3	11.1	47.7	246.2	8.5	13.7	10.8	9.7
	KY	9.2	2/3	347.3	47.7	48.8	17.8	55.6	224.4	15.1	23.2	18.8	11.6
Shatter cones													
Granite	KB	1.8	3/3	319.5	68.1	30.9	22.6	65.8	285.7	31.9	37.9	34.8	14.6
Gneiss	OK	4.5	3/3	350.1	65.2	28.1	23.7	74.1	229.8	31.1	38.4	34.5	15.3
	KE	1.9	3/8	336.4	60.9	15.6	32.4	65.4	250.2	37.8	49.5	43.2	20.5
Mean			8*/23/34	336.0	48.2	17.3	13.7	53.5	241.3	11.8	19.0	14.5	**19.5**
Comp. B													
Granite	KP	8.8	2/2	36.1	70.6			70.1	122.8				
	SJ	3.9	3/3	36.3	56.9	18.0	30.0	56.9	145.4	31.6	43.5	37.1	19.1
Shatter cones													
Granite	KB	1.8	4/5	53.9	55.2	6.8	38.2	47.8	127.5	38.7	54.4	45.9	31.2
	KR	1.8	2/2	50.7	79.1			61.9	67.8				
Gneiss	OK	4.5	3/4	42.3	44.8	52.3	17.2	44.6	146.7	13.7	21.7	17.3	11.2
	JB	1.9	2/2	43.8	61.6			57.9	131.1				
Metavolc. rock	JK	3.4	3/5	55.2	75.2	73.5	9.0	65.0	89.8	15.0	16.4	15.7	9.5
	VN	0	2/5	25.7	67.1			71.3	143.7				
Mean			8*/21/28	42.4	64.1	44.9	8.4	61.0	129.1	10.6	13.3	11.9	12.1
								−61.0	309.1				
Comp. C													
Granite	KK	6.3	1/2	121.8	63.8			27.3	66.8				
Gneiss	KJ	4.2	2/2	178.0	51.7			4.6	26.4				
	VJ	2.1	2/2	204.5	69.4			26.8	8.5				
Shatter cones													
Granite	RK	1.8	1/2	172.8	59.6			12.8	30.3				
Gneiss	OK	4.5	2/4	152.5	58.9			14.2	46.3				
	KE	1.9	1/3	132.8	63.2			23.4	59.3				
Metavolc. rock	HK	3.3	3/3	153.8	75.9	98.4	12.5	37.0	39.1	21.2	23.0	22.1	8.2
Mean			7*12/18	159.5	65.4	32.8	10.7	21.0	39.3	14.1	17.3	15.6	14.1
Comp. E													
Granite	KX	7.2	2/2	258.9	47.1			19.9	317.8				
Granodiorite	VI	22.0	2/2	279.0	28.6			17.6	285.3				
Shatter cones													
Granite	RK	1.8	2/2	302.4	77.3			65.4	328.1				
Gneiss	LK	3.1	2/2	268.4	75.3			51.0	336.8				
	KE	1.9	2/5	278.5	58.0	45.9	37.8	37.3	308.4	40.9	55.6	47.7	12.0
	KI	1.7	3/3	264.7	71.3	19.4	28.8	45.2	332.3	44.0	50.3	47.1	18.4
Metavolc. rock	HK	3.3	1/2	293.8	70.8			56.4	313.5				
Mean			7*/14/18	275.5	62.0	18.4	14.4	39.7	314.3	17.4	22.4	19.7	18.9
								−39.4	134.3				

Notes: *s* = distance from the crater center; *B** = number of sites; *N* = number of samples; *n* = number of specimens; *D* = declination, *I* = inclination; *k* = Fisher's (1953) precision parameter for remanence component, α_{95} = radius of the circle of 95% confidence of the mean pole, Plat and Plong = latitude and longitude of the VGP; *dp* and *dm* = semi-axes of the oval of 95% confidence of the pole; A_{95} = radius of the circle of 95% confidence of the pole; *S* = calculated angular standard deviation between the poles.

magnetic pattern over Keurusselkä. The magnetic signal of pyrrhotite-bearing schist belts is possibly enhanced due to the impact (Raiskila et al. 2011). Similar magnetic highs, indicating increased susceptibilities, for the uplifted rim, have been detected also for the Siljan impact structure in Sweden (Henkel and Aaro 2005).

Target rocks in impact craters are subjected to extreme pressure (up to 100 GPa) and temperature (up to 1000 °C) during the impact event. Typically, the porosity and fracture density decay gradually from the center of the crater to outer regions of the impact structure (Cockell et al. 2005). In the Keurusselkä area,

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

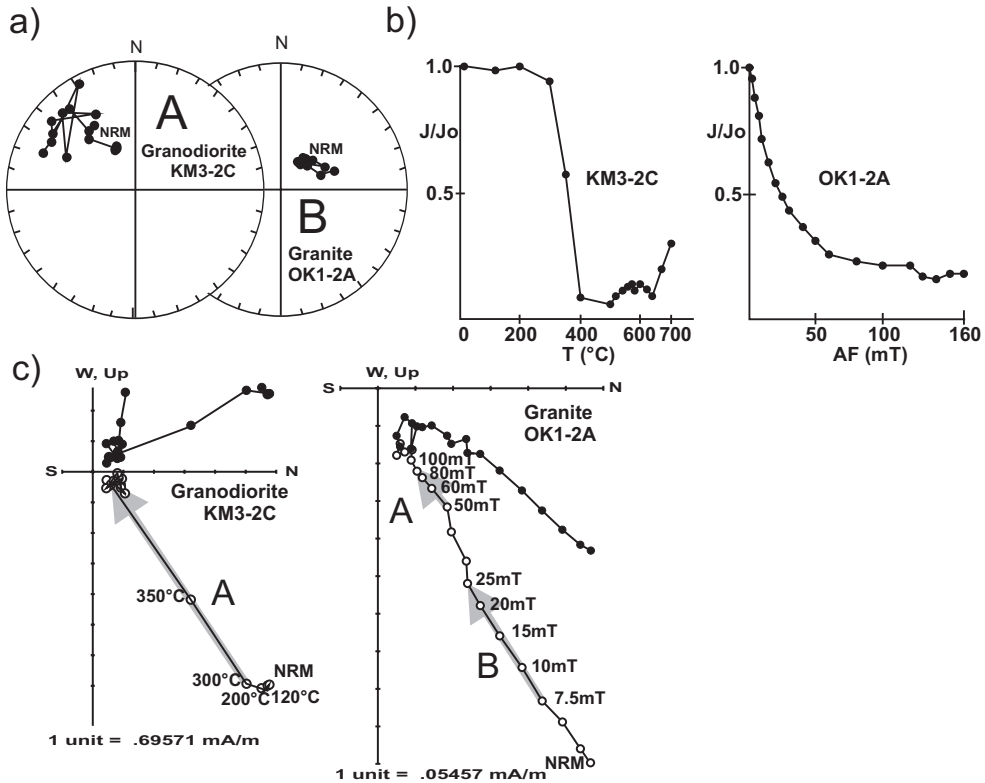


Fig. 8. Examples of the thermal and AF demagnetization behavior for target rock and shatter cone specimens revealing characteristic Svecofennian component A and Mesoproterozoic component B. a) Stereographic projection of component A and B. b) Relative NRM intensity decay of thermal demagnetization for A and AF demagnetization of B. c) Orthogonal vector projections; closed (open) circles denote projections horizontal (vertical) planes.

the average porosities of shatter cones are low because of the weakly shocked crater basement and lack of highly porous impact rocks, such as suevites and impact melt breccias. Average densities of the target rocks are slightly higher than densities of the shatter cones within the same lithologies. The degree of fracturing is associated distinctively with the appearance of shatter cones at the central uplift area, and it is in resemblance to fracturing of the Karikkoselkä impact structure formed on the same granitic target rock (Pesonen et al. 1999).

Carriers of the Magnetization

The carriers for the remanent magnetization in the Keurusselkä rocks, as observed in thermomagnetic and paleomagnetic studies, are magnetite and/or pyrrhotite, which are common and principal carriers in igneous and metamorphic rocks (Dunlop and Özdemir 1997). Demagnetization of the Keurusselkä specimens often shows unstable behavior due to either hydrothermal processes and alteration of magnetite or lack of sufficient amount of magnetic minerals. Acquisition of viscous remanent magnetization (VRM), which typically forms during burial, has disturbed the remanence of low coercivity minerals. Later (impact and postimpact) geological processes have magnetically overprinted the primary remanent magnetization of CFGC. Origin of these secondary remanent magnetizations are either shock (SRM) of the impact event; chemical (CRM), when new minerals are formed or altered in hydrothermal processes; or thermochemical (TCRM) accompanied by the excess heat and fluids of the impact event.

Paleomagnetism

The Neoproterozoic and Phanerozoic evolution of CFGC shows three major tectonic events after crystallization (approximately 1890–1860 Ma): (1) emplacement of rapakivi granites (approximately 1650–1615 Ma), (2) Sveconorwegian orogeny (approximately 1300–900 Ma), and (3) Caledonian orogeny during Silurian-Devonian (approximately 450–350 Ma), when Fennoscandian Shield as a part of Baltica, British Isles, and Laurentia collided (Kohonen and Rämö 2005). Compressional stress during this period may have also

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

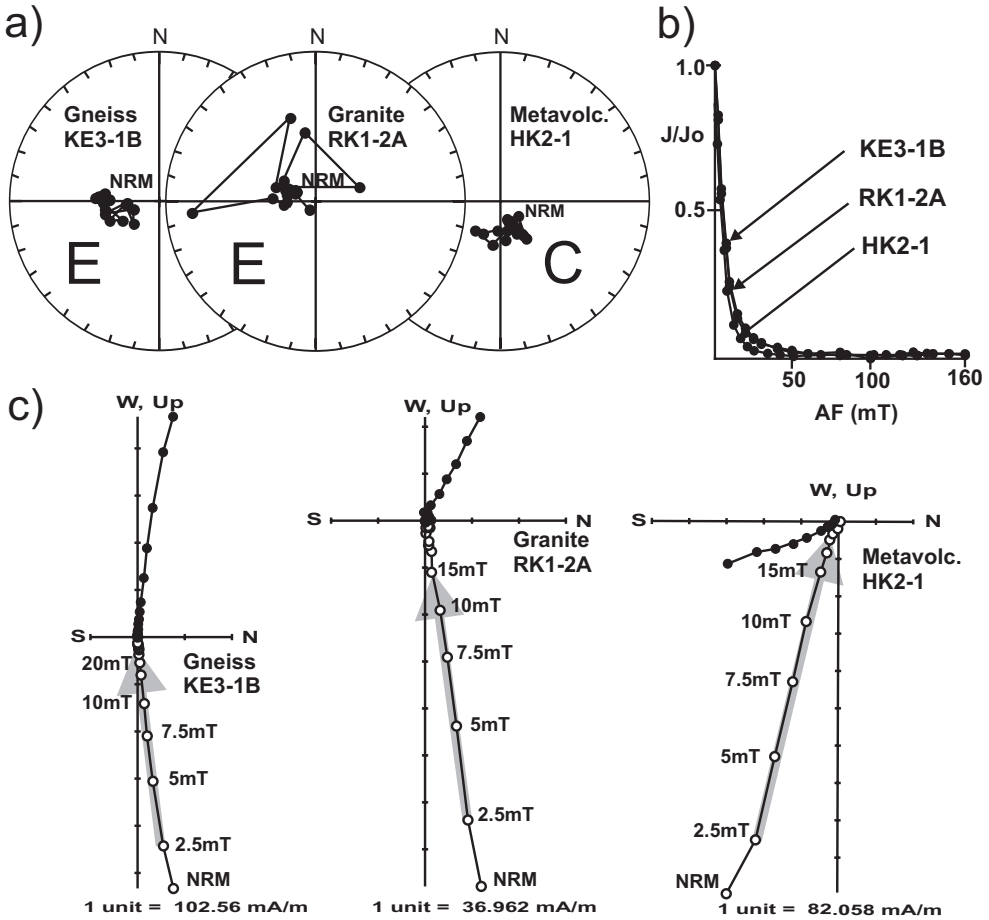


Fig. 9. Examples of the AF demagnetization behavior for shatter cone specimens revealing secondary components C and E. a) Stereographic projections. b) Relative NRM intensity decay of AF demagnetization. c) Orthogonal vector projections; closed (open) circles denote projections horizontal (vertical) planes.

been partly responsible for the formation of shear zones (Mertanen et al. 2008).

The characteristic component A obtained for different lithologies of unshocked target rocks and occasionally from shatter cones shows the typical Svecofennian remanent magnetization direction. This component is interpreted to represent the primary magnetization acquired during the formation and slow cooling of the CFGC at 1880–1860 Ma (U-Pb age for zircon) (Fig. 11A). Field tests for the CFGC area, to confirm the primary nature of magnetization, are not possible. However, this magnetization direction has been widely observed throughout the Fennoscandian Shield (e.g., see Neuvonen et al. 1981; Elming 1985; Pesonen et al. 1991; Mertanen and Pesonen 1994), and this corresponds also with the component A, reported by Mertanen and Pesonen (1995) and Pesonen et al. (1999) for the unshocked granitic target rock from Karikkoselkä impact structure.

Three secondary components B, C, and E are isolated. Component B is obtained from shatter cones yielding a pole (Plat = 60.1°, Plon = 122.8°, A_{95} = 12.6°). This pole does not agree with the Precambrian key poles of Baltica (Elming et al. 1993). However, it is close to the pole of a well-dated (1122 ± 7 Ma) Salla diabase dyke (Salminen et al. 2009) (Fig. 11A), indicating late Mesoproterozoic magnetization age. This agrees with the U-Pb age of 1140 ± 6 Ma of the pseudotachylitic breccia vein (Schmieder et al. 2009), supporting the idea that component B represents the magnetization due to the impact event. Component B has also low coercivity, which could be an indication of shock remanent magnetization (Halls 1979; Elming and Bylund 1991). Keurusselkä and Salla poles are far from the Precambrian APWP of Baltica, thus creating a large clockwise loop from 1265 Ma to 1036 Ma, Salla pole (1122 Ma) being the apex of the loop. Salminen et al. (2009) discussed the similarities between the

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

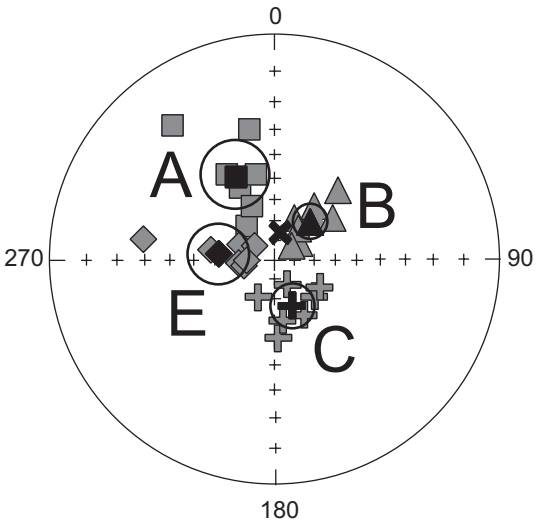


Fig. 10. Site mean paleomagnetic directions for target rock and shatter cones. Component A refers to characteristic Svecofennian direction. Components B, C, and E denote the secondary directions isolated from specimens. Black symbols display site means with α_{95} confidence about the means and gray symbols overall mean directions. Present Earth Field direction is marked as black X.

pre-Sveconorwegian (approximately 1.3–1.0 Ga) APWP loops of Baltica, Laurentia (including Logan Loop), and Kalahari-Grunehogna. However, these cratons seem to reach their apexes at different times. Yet, it is evident that the APWPs of Baltica, Laurentia, and Kalahari express similar loop-like shapes, suggesting that they may share a similar history. Salminen et al. (2009) proposed that Baltica and Laurentia drifted together until 1.12 Ga, but as Pesonen et al. (2003) showed, they separated at 1.04 Ga. Therefore, the more likely explanation for the observed large loops is the proposed true polar wander (Evans 2003) at the late Mesoproterozoic. However, dated paleomagnetic poles for approximately 1.25–1.12 Ga interval for these continents are required to test the similarity. Nevertheless, the poles from the Keurusselkä impact structure and from the Salla dyke provide hints that the proposed Mesoproterozoic Baltica–Laurentia unity in the Columbia (Hudsonland, Nuna) supercontinent assembly may have lasted until 1.12 Ga (see also Salminen et al. 2009).

The opposite polarity of Keurusselkä pole B fits also to the Phanerozoic APW path (230 Ma) (Fig. 11B). Mertanen and Pesonen (1995), Mertanen et al. (2008), and Preeden et al. (2009) have presented similar paleopoles (265 Ma) for paleomagnetic data obtained from South Finland shear zones. Mertanen et al. (2008) and Preeden et al. (2009) favored Phanerozoic age for the paleopole and interpreted that this remagnetization was caused by oxidizing fluids.

Paleopole C, when plotted on the Phanerozoic APWP curve, is close to an age 460–470 Ma, which relates to the Caledonian orogeny.

Paleopole E, however, does not fit to the Phanerozoic APW path. Previously, based on a positive conglomerate test, primary remanence direction corresponding to pole E of this study has been obtained from 2058 ± 6 Ma Kuetsyarvi formation in northern Pechenga zone in the Kola Province (Torsvik and Meert 1995; Melezhik et al. 2007). Moreover, similar but dual polarity component was also seen as an overprint in the 2.44 Ga Koillismaa-layered intrusions E(comb) (D = 261°, I = 51°, α_{95} = 10°, Plat = 23° N, Plon = 327° E, A_{95} = 11°) (Mertanen et al. 1989). They describe that it was magnetically hardest and isolated at blocking temperatures of 680 °C being of thermochemical origin. This component was, however, observed only in few samples right at the contact of the dyke and in basement rocks. Mertanen et al. (1989) proposed that component E resulted from Jatulian rifting, which fractured the crust about 2200–2000 Ma ago. Furthermore, similar component is isolated for four mafic dykes and their baked basement rocks located close by NW-SE trending mega shear zone in Finnish Karelia at the Taivalkoski area (Salminen et al. Forthcoming). These dykes have not been dated, but the corresponding pole plotted on the apparent polar wander path (APWP) of Baltica indicates an age of 2.1–2.06 Ga. For the Keurusselkä area, this age is not possible, as the basement rocks are of Svecofennian age. Interestingly, the south pole of opposite polarity of Keurusselkä component E resembles a pole position for Winter Coast formation reported by Popov et al. (2002), which differs greatly from Phanerozoic APWP reported by Torsvik et al. (1992, 1996), and is not given further attention here.

However, these observations are important as they will provide more perspectives for discussion of the origin of different magnetization components observed throughout the Fennoscandian Shield.

CONCLUSIONS

The Keurusselkä structure is a deeply eroded complex crater lacking the impact melt and rock lithologies, such as suevite and breccia. Impact features include remarkable shatter cones in crystalline basement rocks with distinct planar deformation features (PDFs) in quartz. Moreover, the basement rocks are heavily fractured due to the impact shock.

Petrophysical and rock magnetic properties imply that increased magnetization and susceptibility within the area, where shatter cones were observed, represent the central uplift with diameter of 5–10 km. The main magnetic minerals identified from thermomagnetic,

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

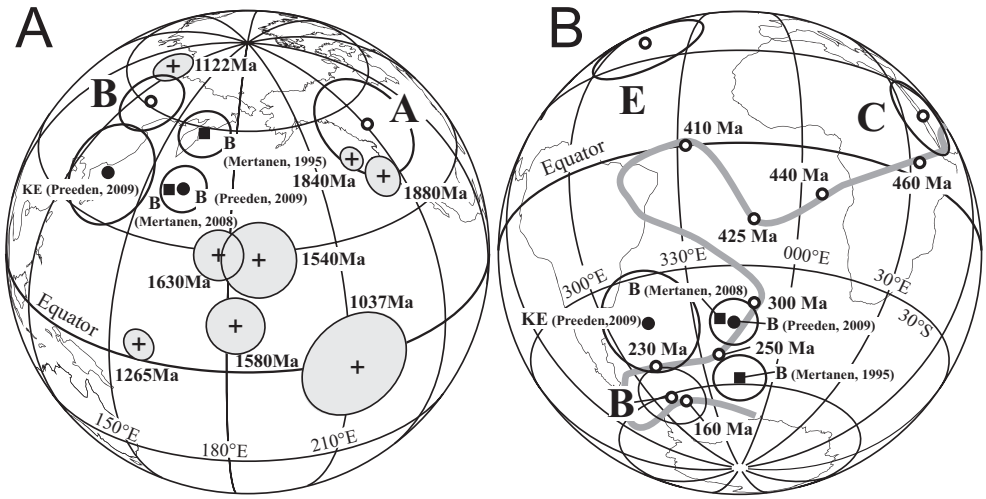


Fig. 11. A) Mean paleomagnetic poles for characteristic Svecofennian component A and secondary component B calculated from shatter cone sites and unshocked target rock sites. Baltica "key poles" and chosen well-dated poles are marked as crosses with A_{95} circles (Buchan et al. 2000; Salminen et al. 2009). B) Fennoscandian Phanerozoic APW path (Torsvik et al. 1996; Smethurst et al. 1998; Torsvik and Rehnström 2001, 2003) and mean paleomagnetic poles for the south pole of component B and secondary components C and E.

hysteresis, and SEM analysis are magnetite and pyrrhotite. The pyrrhotite-bearing schist belts, cutting the central uplift, are proposed to be enhanced due to the impact. Alteration of minerals in hydrothermal processes in the crater basement could be the reason for increased magnetic properties.

Four paleomagnetic components for the Keurusselkä rocks are interpreted to record the regional and global Mesoproterozoic and Paleozoic geological events during the evolution of the Svecofennian Shield. Based on the U-Pb (Zr) age of the granitoids and the similarity with the other Svecofennian remanent magnetization directions, the component A of this study is interpreted to represent the primary magnetization of the Keurusselkä rocks being acquired during the Svecofennian orogeny (1890–1840 Ma). Components B, C, and E represent remagnetizations in target rock, crater basement, and shatter cones. Component B is interpreted to represent magnetization related to the impact event based on its agreement with the 1122 ± 7 Ma virtual geomagnetic pole of the Salla diabase dyke and the $^{40}Ar/^{39}Ar$ age of the structure (1140 ± 6 Ma). This new pole from the Keurusselkä impact structure gives further support for the proposed large APWP loop of Baltica before Sveconorwegian (Salminen et al. 2009). Moreover, also APWPs of Laurentia and Kalahari express similar large loops. As Baltica and Laurentia have been proposed to drift together until 1.12 Ga, but separating after that (Pesonen et al. 2003), a common rapid drift is not explaining observed large loops. More likely explanation for observed loops is the true polar wander. Components C and E are interpreted to represent remagnetizations, which are related to younger geological events on the area and are considered too young for the impact event.

Acknowledgments—We thank Dr. Agnes Kontny and Dr. Jüri Plado for thorough and constructive reviews. We also thank Dr. Martti Lehtinen for his valuable help on this research project. Dr. Jüri Plado (University of Tartu), Dr. Fabio Donadini (ETH Zürich), Dr. Teemu Öhman (LPI, Houston), Jonna Poikolainen (Univ. of Oulu), and M.Sc. Robert Klein (University of Helsinki) are thanked for participating in field trips. We thank the following people of Geological Survey of Finland: Dr. Bo Johansson and Dr. Marja Lehtonen for the help with SEM, Hanna Leväniemi for providing aeromagnetic map, and Heikki Säävuori for providing petrophysical data from GSF database.

This research project is funded by the Academy of Finland. S. Raiskila acknowledges the Finnish Graduate School in Geology and the Vilho, Yrjö and Kalle Väisälä foundation for supporting funds. Furthermore, the Academy of Finland, Finnish Cultural foundation, Emil Aaltonen foundation, and Oskar Malm foundation are thanked for supporting funds for J. Salminen. The Vilho, Yrjö and Kalle Väisälä foundation is thanked for financial support for T. Elbra.

Editorial Handling—Dr. Christian Koeberl

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

REFERENCES

Abels A., Plado J., Pesonen L. J., and Lehtinen M. 2002. The impact cratering record of Fennoscandia. In *Impacts in Precambrian shields*, edited by Plado J. and Pesonen L. J. Berlin: Springer Verlag. pp. 1–58.

Buchan K., Mertanen S., Elming S-Å., Park R. G., Pesonen L. J., Bylund G., and Abrahamsen N. 2000. The drift of Laurentia and Baltica in the Proterozoic: A comparison based on key paleomagnetic poles. *Tectonophysics* 319:167–198.

Cockell C. S., Lee P., Broady P., Lim D. S. S., Osinski G. R., Parnell J., Koeberl C., Pesonen L. J., and Salminen J. 2005. Effects of asteroid and comet impacts on habitats for lithophytic organisms—A synthesis. *Meteoritics & Planetary Science* 40:1901–1914.

Dekkers M. J. 1988. Magnetic properties of natural pyrrhotite. Part I: Behaviour of initial susceptibility and saturation magnetization related rock magnetic parameters in a grain size dependent framework. *Physics of the Earth and Planetary Interiors* 52:376–393.

Dunlop D. J. and Özdemir Ö. 1997. *Rock magnetism, fundamentals and frontiers*. Cambridge, UK: Cambridge University Press. 573 p.

Dypvik H., Plado J., Heinberg C., Håkansson E., Pesonen L. J., and Raiskila S. 2008. Impact structures and events—A Nordic perspective. *Episodes* 31:107–114.

Earth Impact Database. 2011. http://www.unb.ca/passc/ImpactDatabase/. Accessed September 10, 2011.

Elming S.-Å. 1985. A paleomagnetic study of the Svecokarelian basic rocks from northeastern Sweden. *Geologiska Föreningens i Stockholm Förhandlingar* 107:17–35.

Elming S.-Å. and Bylund G. 1991. Paleomagnetism and the Siljan impact structure, central Sweden. *Geophysical Journal International* 105:757–770.

Elming S.-Å., Pesonen L. J., Leino M. A. H., Khramov A. N., Mikhailova N. P., Krasnova A. F., Mertanen S., Bylund G., and Terho M. 1993. The drift of Fennoscandian and Ukrainian shields during the Precambrian: A paleomagnetic analysis. *Tectonophysics* 223:177–198.

Evans D. A. D. 2003. True polar wander and supercontinents. *Tectonophysics* 362:303–320.

Ferrière L., Morrow J. R., Amgaa T., and Koeberl C. 2009. Systematic study of universal-stage measurements of planar deformation features in shocked quartz: Implications for statistical significance and representation of features. *Meteoritics & Planetary Science* 44:925–940.

Ferrière L., Raiskila S., Osinski G., Pesonen L. J., and Lehtinen M. 2010. The Keurusselkä impact structure, Finland—Impact origin confirmed by characterization of planar deformation features in quartz grains. *Meteoritics & Planetary Science* 45:434–446.

Fisher R. 1953. Dispersion of a sphere. *Proceedings of the Royal Society of London* 217:295–305.

French B. M. 1998. *Traces of catastrophe: A handbook of shock-metamorphic effects in terrestrial meteorite impact structures*. LPI Contribution 954. Houston, Texas: Lunar and Planetary Institute. 120 p.

French B. M. and Koeberl C. 2010. The convincing identificaton of terrestrial meteorite impact structures: What works, what doesn't, and why. *Earth Science Reviews* 98:123–170.

Halls H. 1979. The Slate Island meteorite impact site: A study of shock remanent magnetization. *Geophysical Journal of Royal Astronomical Society* 59:553–591.

Henkel H. and Aaro S. 2005. Geophysical investigations of the Siljan impact structure—A short review. In *Impact tectonics*, edited by Koeberl C. and Henkel H. Berlin: Springer Verlag. pp. 247–288.

Henkel H. and Pesonen L. J. 1992. Impact craters and craterform structures in Fennoscandia. *Tectonophysics* 216:31–30.

Hietala S. and Moilanen J. 2004. Keurusselkä—A new impact structure in central Finland (abstract #1619). 35th Lunar and Planetary Science Conference. CD-ROM.

Hietala S. and Moilanen J. 2007. Keurusselkä—distribution of shatter cones (abstract #1762). 38th Lunar and Planetary Science Conference. CD-ROM.

Kähkönen Y. 2005. Svecofennian supracrustal rocks. In *Precambrian geology of Finland—Key to the evolution of the Fennoscandian shield*, edited by Lehtinen M., Nurmi P. A., and Rämö O. T. Developments in Precambrian Geology, Amsterdam: Elsevier. vol. 14. 736 p.

Kirschvink J. L. 1980. The least-squares line and plane and the paleomagnetic data. *Geophysical Journal of the Royal Astronomical Society* 62:699–718.

Kivekäs L. 1993. Density and porosity measurements at the petropysical laboratory of the Geological Survey of Finland. In *Geological survey of Finland, current research 1991–1992*, edited by Autio S. GSA Special Paper 18. Espoo: Geological Survey of Finland. pp. 119–127.

Kohonen J. and Rämö O. T 2005. Sedimentary rocks, diabases, and late cratonic evolution. In *Precambrian geology of Finland—Key to the evolution of the Fennoscandian Shield*, edited by Lehtinen M., Nurmi P. A., and Rämö O. T. Developments in Precambrian Geology, Amsterdam: Elsevier. vol. 14. 736 p.

Lehtinen M. 1976. Lake Lappajärvi, a meteorite impact site in western Finland. *Geological Survey of Finland Bulletin* 285:1–92.

Leino M. A. H. 1991. *Multicomponent analysis of paleomagnetic data by least-squares method*. Report Q29.1/91/2. Espoo: Geological Survey of Finland. 15p. In Finnish.

Melezhik V. A., Huhma H., Condon D. J., Fallick A. E., and Whitehouse M. J. 2007. Temporal constraints on the Paleoproterozoic Lomagundi-Jatuli carbon isotopic event. *Geology* 35:655–658.

Melosh H. J. 1989. *Impact cratering: A geologic process*. Oxford Monographs on Geology and Geophysics, New York: Oxford University Press. vol. 11. 253 p.

Mertanen S. and Pesonen L. J. 1994. Preliminary results of a paleomagnetic and rockmagnetic study of the Proterozoic Tsuomasvarri intrusions, northern Fennoscandia. *Precambrian Research* 69:25–50.

Mertanen S. and Pesonen L. J. 1995. Paleomagnetic and rock magnetic investigations of the Sipoo Subjotnian quartz porphyry and diabase dykes, southern Finland. *Physics of the Earth and Planetary Interiors* 88:145–175.

Mertanen S., Pesonen L. J., Huhma H., and Leino M. A. H. 1989. Paleomagnetism of the Early Proterozoic layered intrusions, northern Finland. *Geological Survey of Finland Bulletin* 347:1–40.

Mertanen S., Airo M.-L., Elminen T., Niemelä R., Pajunen M., Wasenius P., and Wennerström M. 2008. Paleomagnetic evidence for Mesoproterozoic-Paleozoic reactivation of the Paleoproterozoic crust in Southern Finland. *Geological Survey of Finland, Special Paper* 47:215–252.

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Neuvonen K. J., Korsman K., Kouvo O., and Paavola J. 1981. Paleomagnetism and age relations of the rocks in the main sulphide ore belt in central Finland. *Geological Survey of Finland Bulletin* 53:109–133.

Nironen M. 2003. *Central Finland Granitoid complex—Explanation to a map. Report of investigation 157*. Espoo: Geological Survey of Finland. 45 p.

O'Reilly W. 1984. *Rock and mineral magnetism*. Glasgow: Blackie. 220 p.

Palmu J. P. 1982. Paleomagneettinen menetelmä eroosiopinnan muinaisen syvyyden määrittämiseksi (Paleomagnetic method to estimate the erosion level). M.Sc. thesis. Department of Mining and Metallurgy, Helsinki Technical University (TKK), Espoo, Finland. In Finnish.

Pesonen L. J., Torsvik T. H., Elming S-Å., and Bylund G. 1989. Crustal evolution of Fennoscandia—Paleomagnetic constrains. *Tectonophysics* 162:27–49.

Pesonen L. J., Bylund G., Torsvik T. H., Elming S-Å., and Mertanen S. 1991. Catalogue of paleomagnetic directions and poles from Fennoscandia: Archaean to tertiary. *Tectonophysics* 195:151–207.

Pesonen L. J., Marcos N., and Pipping F. 1992. Palaeomagnetism of the Lappajärvi impact structure, western Finland. *Tectonophysics* 216:123–142.

Pesonen L. J., Elo S., Lehtinen M., Jokinen T., Puranen R., and Kivekäs L. 1999. Lake Karikkoselkä impact structure, central Finland: New geophysical and petrographic results. In *Large meteorite impacts and planetary evolution II*, edited by Dressler B. O. and Sharpton V. L. GSA Special Paper 339. Boulder, Colorado: Geological Society of America. pp. 131–139.

Pesonen L. J., Elming S-Å., Mertanen S., Pisarevsky S., D'Agrella-Filho M. S., Meert J. G., Schmidt P. W., Abrahamsen N., and Bylund G. 2003. Paleomagnetic configuration of continents during the Proterozoic. *Tectonophysics* 375:289–324.

Pesonen L. J., Poutanen M., and Ruotsalainen H. E. 2004. The Keurusselkä impact strucuture, Central Finland—Preliminary geophysical data (abstract #5068). *Meteoritics & Planetary Science* 39.

Pilkington M. and Grieve R. A. F. 1992. Geophysical signatures of terrestrial impact craters. *Reviews of Geophysics* 30:161–181.

Pipping F. and Lehtinen M. 1992. Geology, stratigraphy and structure of the Lappajärvi meteorite crater, western Finland: Preliminary results of deep drilling. *Tectonophysics* 216:91–97.

Plado J., Pesonen L. J., Elo S., Puura V., and Suuroja K. 1996. Geophysical research on the Kärdla impact structure, Hiiumaa Island, Estonia. *Meteoritics & Planetary Science* 31:289–298.

Popov V., Iosifidi A., and Khramov A. 2002. Paleomagnetism of Upper Vendian sediments from the Winter Coast, White Sea region, Russia: Implications for the paleography of Baltica during Neoproterozoic times. *Journal of Geophysical Research* 107:2315–2323.

Preeden U., Mertanen S., Elminen T., and Plado J. 2009. Secondary magnetization in shear and fault zones in southern Finland. *Tectonophysics* 479:203–213.

Puranen R., Pekkarinen L. J., and Pesonen L. J. 1992. Interpretation of magnetic fabrics in the Early Proterozoic diabase dykes of Keuruu, central Finland. *Physics of the Earth and Planetary Interiors* 72:68–82.

Raiskila S, Preeden U., Heikkilä P., Elbra T., and Pesonen L. J. 2011. Physical properties of Vilppula drill cores and petrographic analysis of associated breccias in Keurusselkä impact structure, central Finland. *Studia Geophysica et Geodaetica* 56.

Reimold W. U. 1998. Exogenic and endogenic breccias: A discussion of major problematics. *Earth Science Reviews* 43:25–47.

Reimold W. U., Kelley S. P., Sherlock S. C., Henkel H., and Koeberl C. 2005. Laser Argon dating of melt impact breccias from Siljan impact structure, Sweden: Implications for a possible relationship to Late Devonian extinction events. *Meteoritics & Planetary Science* 40:591–607.

Roberts A. P., Pike C. R., and Verosu L. 2000. First-order reversal curve diagrams: A new tool for characterizing the magnetic properties of natural samples. *Journal of Geophysical Research* 105:461–475.

Ruotsalainen H. E., Hietala S., Poutanen M., Moilanen J., and Pesonen L. J. 2006. Gravity survey of the Keurusselkä meteorite impact structure. *Bulletin of the Geological Society of Finland*, Special Issue 1, 27th Nordic Geological Winter Meeting, January 9–12, 2006, Oulu, Finland. 134 p.

Säävuori H. and Hänninen R. 1997. Finnish petrophysical database. Proceeedings, Petrophysics in potential field interpretation: First Workshop for the Finnish Geophysical Crustal Model Program, Geological Survey of Finland.

Salminen J., Donadini F., Pesonen L. J., Masaitis V. L., and Naumov M. V. 2006. Paleomagnetism and petrophysics of the Jänisjärvi impact structure, Russian Karelia. *Meteoritics & Planetary Science* 41:1853–1870.

Salminen J., Pesonen L. J., Mertanen S., Vuollo J., and Airo M-L. 2009. Paleomagnetism of Salla Diabase Dyke, northeastern Finland and its implication to the Baltica-Laurentia entity during Mesoproterozoic. In *Paleoproterozoic supercontinents and global evolution*, edited by Reddy S. S., Mazumber R., and Evans D. A. D. London: Geological Society of London, Special Publications, 232. pp. 199–217.

Salminen J., Mertanen S., Halls H. C., Pesonen L. J., and Vuollo J. Forthcoming. Paleomagnetic and rock magnetic studies on the 2.45–2.1 Ga Diabase Dykes of Karelia, East Finland—Key for testing the proposed superia supercraton. *Precambrian Research*.

Schmieder M., Jourdan F., Hietala S., Moilanen J., Öhman T., and Buchner E. 2009. A high—precision late Mesoproterozoic $^{40}Ar/^{39}Ar$ age for the Keurusselkä impact structure (Finland) (abstract #1028). 40th Lunar and Planetary Science Conference. CD-ROM.

Smethurst M. A., Krhamov A. N., and Pisarevsky S. 1998. Paleomagnetism of the Lower Ordovician Orthoceras Limestone, St. Petersburg, and revised drift history for Baltica in the Early Palaeozoic. *Geophysical Journal International* 133:44–56.

Törnberg R. and Sturkell E. F. F. 2005. Density and magnetic susceptibility of rocks from the Lockne and Tvären marine impact structures. *Meteoritics & Planetary Science* 40:639–651.

Torsvik T. H. and Meert J. G. 1995. Early Proterozoic paleomagnetic data from the Pechenga zone (north-west Russia) and their bearing on Early Proterozoic palaeography. *Geophysical Journal International* 122:520–536.

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Torsvik T. H. and Rehnström E. F. 2001. Cambrian palaeomagnetic data from Baltica: Implications for true polar wander and Cambrian palaeogegraphy. *Journal of the Geological Society* 158:321–329.

Torsvik T. H. and Rehnström E. F. 2003. The Tornquist Sea and Baltica-Avalonia docking. *Tectonophysics* 362:67–82.

Torsvik T. H., Smethurst M. A., Van der Voo R., Trench A., Abrahamssen N., and Halvorsen E. J. 1992. BALTICA: A synopsis of Vendian-Permian palaeomagnetic data and their palaeotectonic implications. *Earth Science Reviews* 33:133–152.

Torsvik T. H., Smethurst M. A., Meert J. G., Van der Voo R., McKerrow W. S., Brasier M. D., Strut B. A., and Walderhaug H. J. 1996. Continental breakup and collision in the Neoproterozoic and Paleoproterozoic—A tale of Baltica and Laurentia. *Earth Science Reviews* 40:229–258.

Tsikalas F., Gudlaugsson S. T., Eldholm O., and Faleide J. I. 1998. Integrated geophysical analysis supporting the impact origin of the Mølnir structure, Barents Sea. *Tectonophysics* 289:257–280.

Wehland F., Stancu A., Rochette P., Dekkers M. J., and Appel E. 2005. Experimental evaluation of magnetic interaction on pyrrhotite-bearing samples. *Physics of the Earth and Planetary Interiors* 153:181–190.

Winklhofer M. and Zimanyi G. T. 2006. Extracting the intrinsic switching field distribution in perpendicular media: A comparative analysis. *Journal of Applied Physics* 99:08E710-1-3.

Zijderveld J. D. 1967. A.C. demagnetization in rocks: Analysis of results. In *Methods in paleomagnetism*, edited by Collinson D. W., Creer K. M., and Runcorn S. K. New York: Elsevier. pp. 254–286.

19455100, 2011, 11, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1945-5100.2011.01255.x by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License